Ex No: Content Beyond Syllabus - 1: POSTFIX EXPRESSION EVALUATION USING STACK

PROGRAM

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
#include <stdlib.h>

struct Stack {
    int top;
    unsigned capacity;
    int* array;
};

struct Stack* createStack(unsigned capacity) {
    struct Stack* stack = (struct Stack*) malloc(sizeof(struct Stack));
    if (!stack) return NULL;
    stack->top = -1;
    stack->capacity = capacity;
    stack->array = (int*) malloc(stack->capacity * sizeof(int));
    if (!stack->array) return NULL;
    return stack;
}

int isEmpty(struct Stack* stack) {
    return stack->top == -1;
}

char peek(struct Stack* stack) {
    return stack->array[stack->top];
}

char pop(struct Stack* stack) {
    if (!isEmpty(stack))
        return stack->array[stack->top--];
    return '$';
}

void push(struct Stack* stack, char op) {
    stack->array[++stack->top] = op;
}

int evaluatePostfix(char* exp) {
    struct Stack* stack = createStack(strlen(exp));
    int i;
    if (!stack) return -1;
    for (i = 0; exp[i]; ++i) {
        if (isdigit(exp[i]))
            push(stack, exp[i] - '0');
        else {
            int val1 = pop(stack);
            int val2 = pop(stack);
```

```
            switch (exp[i]) {
                case '+': push(stack, val2 + val1); break;
                case '-': push(stack, val2 - val1); break;
                case '*': push(stack, val2 * val1); break;
                case '/': push(stack, val2 / val1); break;
            }
        }
    }
    return pop(stack);
}

int main() {
    char exp[] = "231*+9-";
    printf("postfix evaluation: %d", evaluatePostfix(exp));
    return 0;
}
```